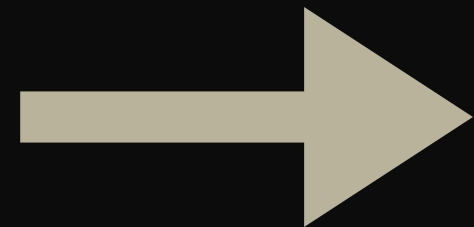


EDUCATION & ECONOMICS IN KYRGYZSTAN

DATASET

STAT.GOV.KG



.xls

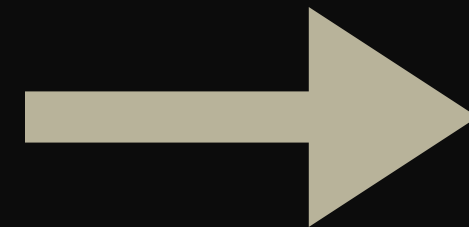


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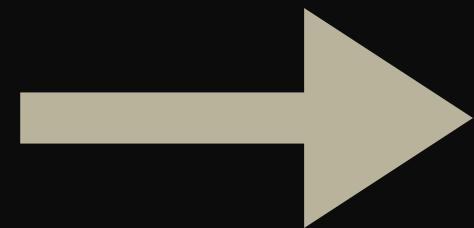


.xls

...



01-cleaning.ipynb



FINAL_DATASET.CSV

DATASET



**14 OBSERVATIONS (2011-2024),
33 COLUMNS**

**6 MAJOR ECONOMIC INDICATORS
(GDP, SALARIES, UNEMPLOYMENT, ETC.)**

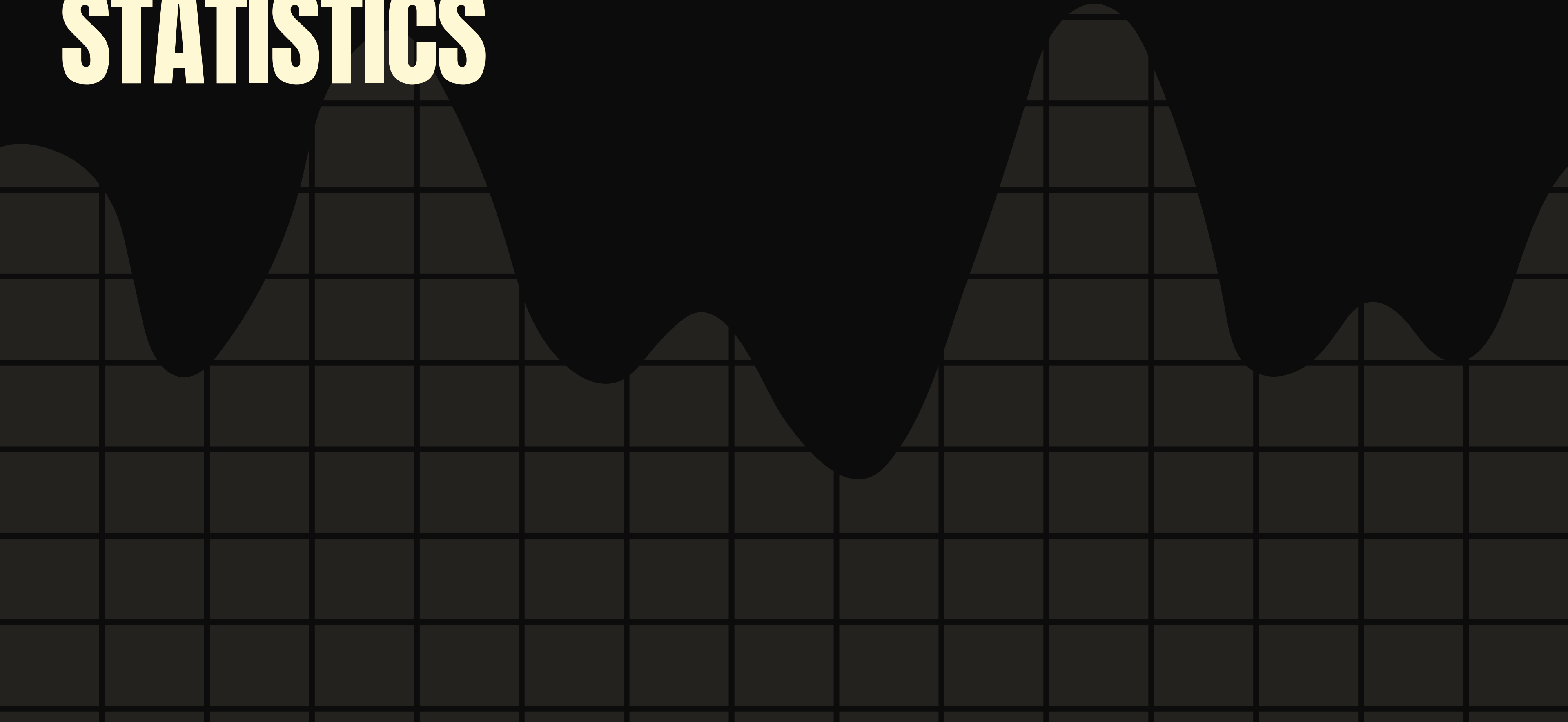
**4 MAJOR EDUCATION INDICATORS
(EDUCATIONAL FACILITIES, TEACHERS,
GRADUATES, STATE EXPENDITURES)**

RESEARCH QUESTIONS

- How have key education and economy indicators in Kyrgyzstan changed between 2011 and 2024?
- Is there a statistically significant association between education indicators and economy indicators in Kyrgyzstan?
- Can education spending be considered a direct cause of economic growth in Kyrgyzstan?
- What is the most plausible explanation for the relationship between education and economic development in Kyrgyzstan?

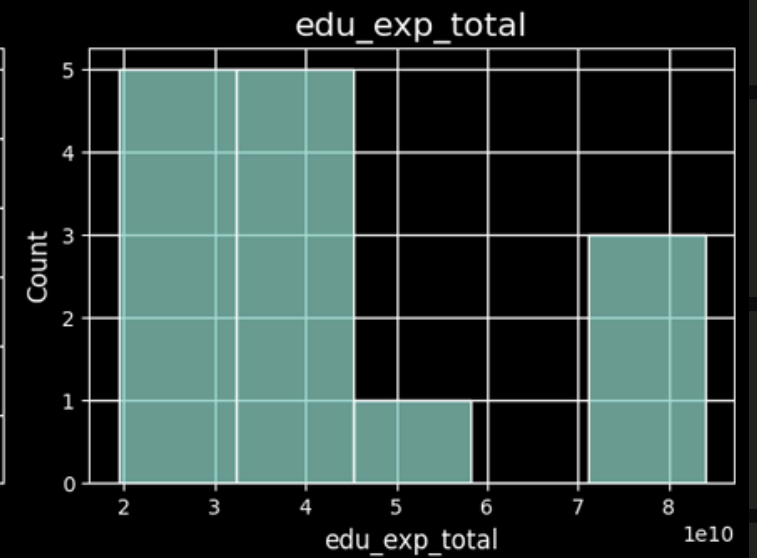
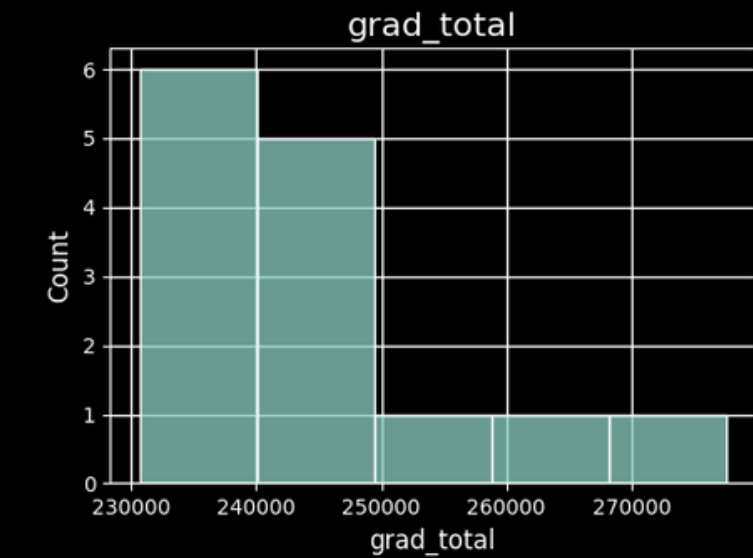
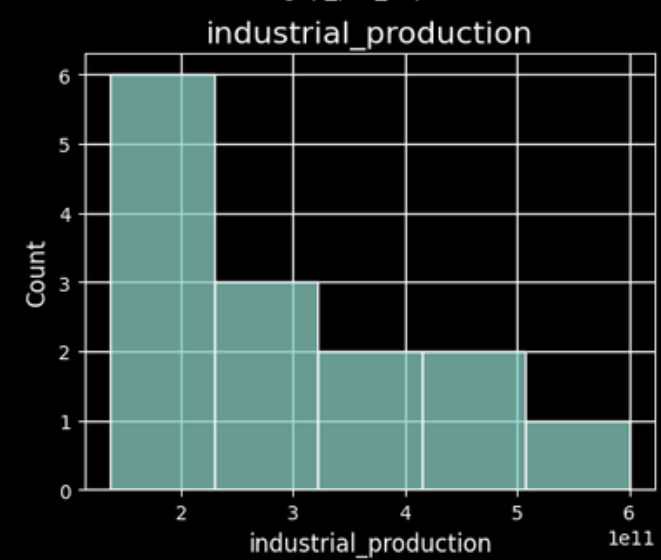
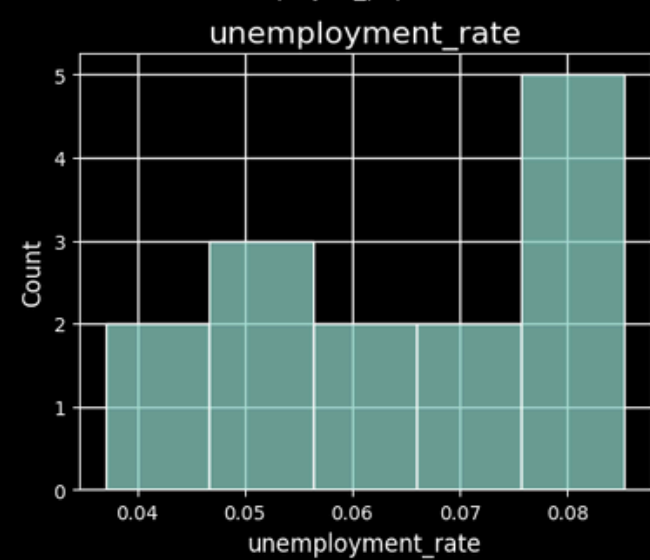
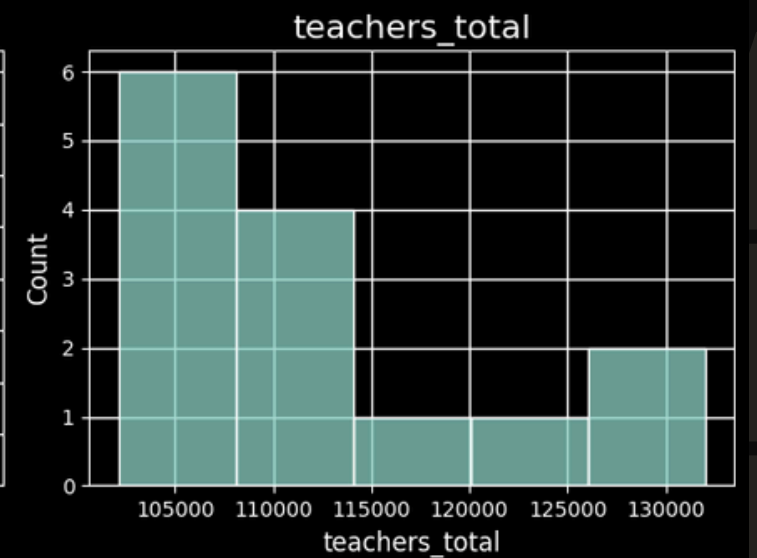
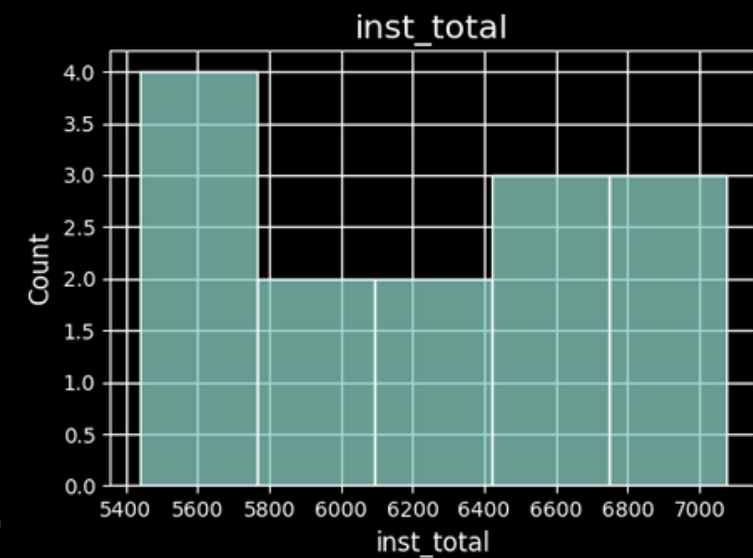
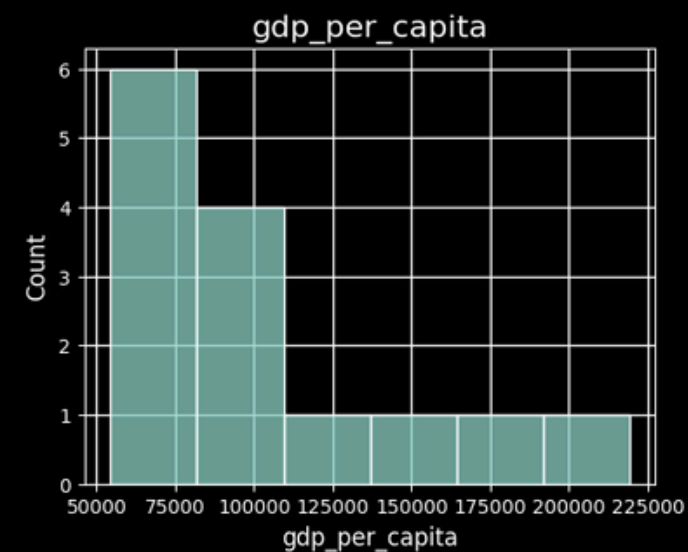
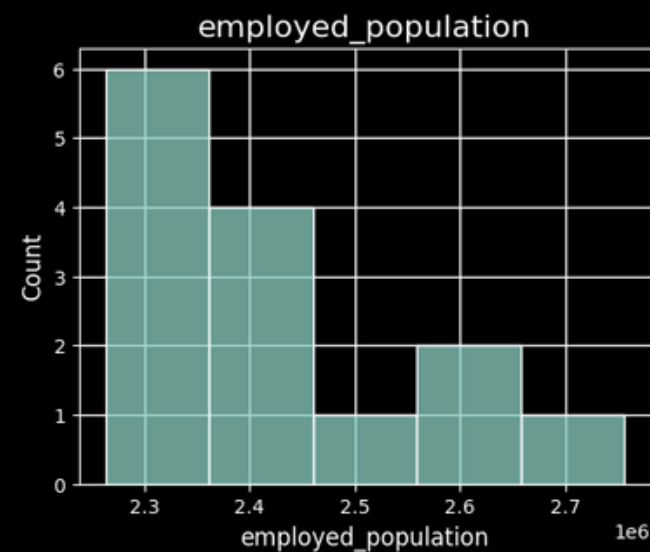
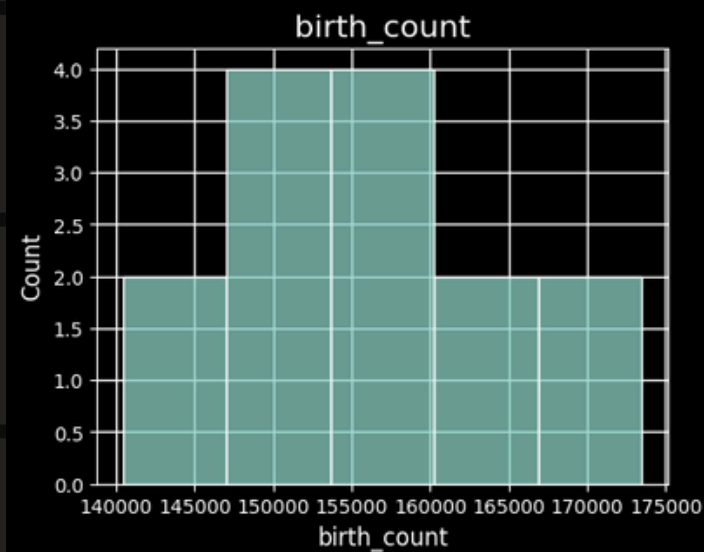


DESCRIPTIVE STATISTICS



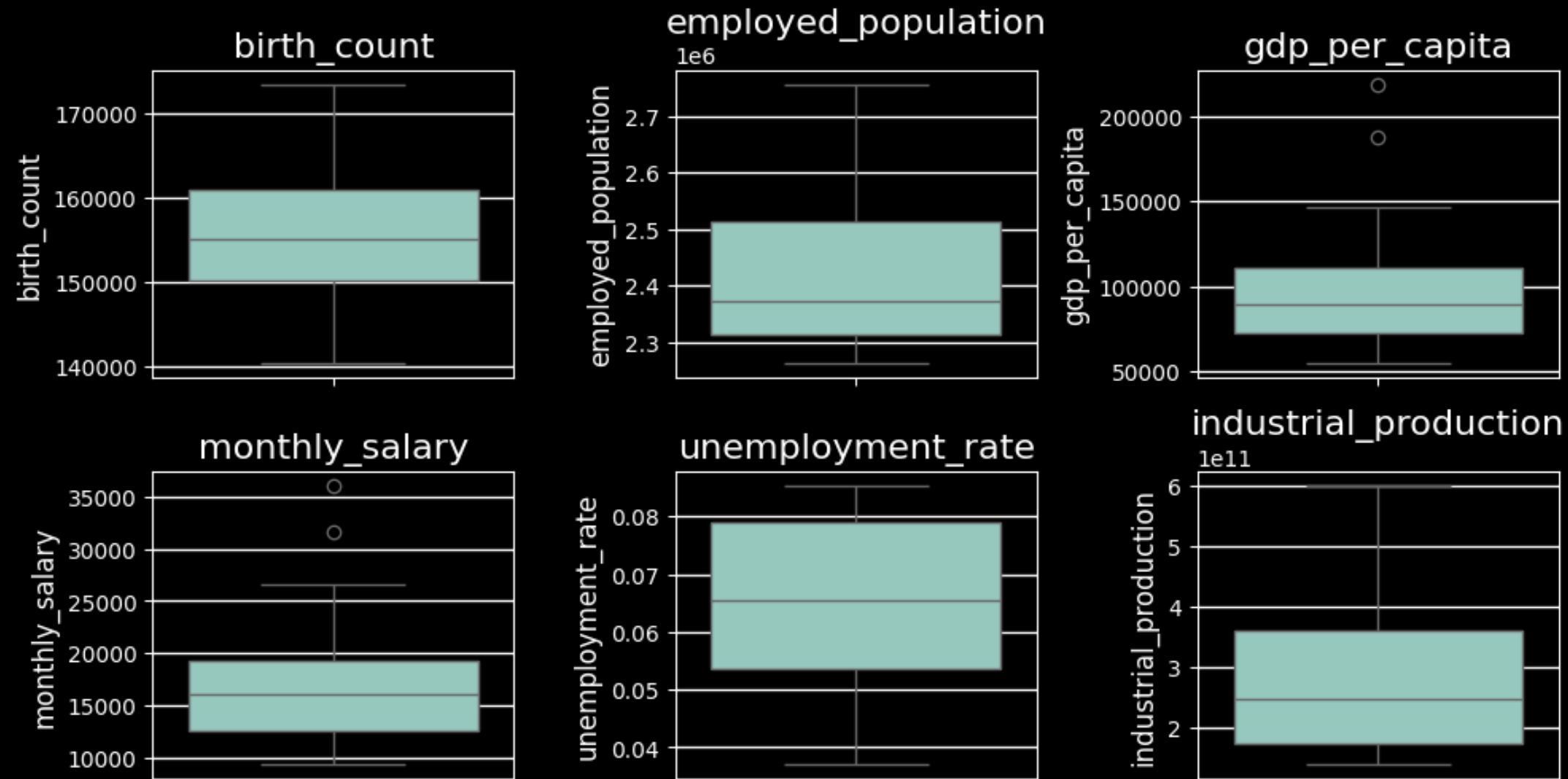
HISTOGRAMS

Distribution of Economy Variables: Histograms

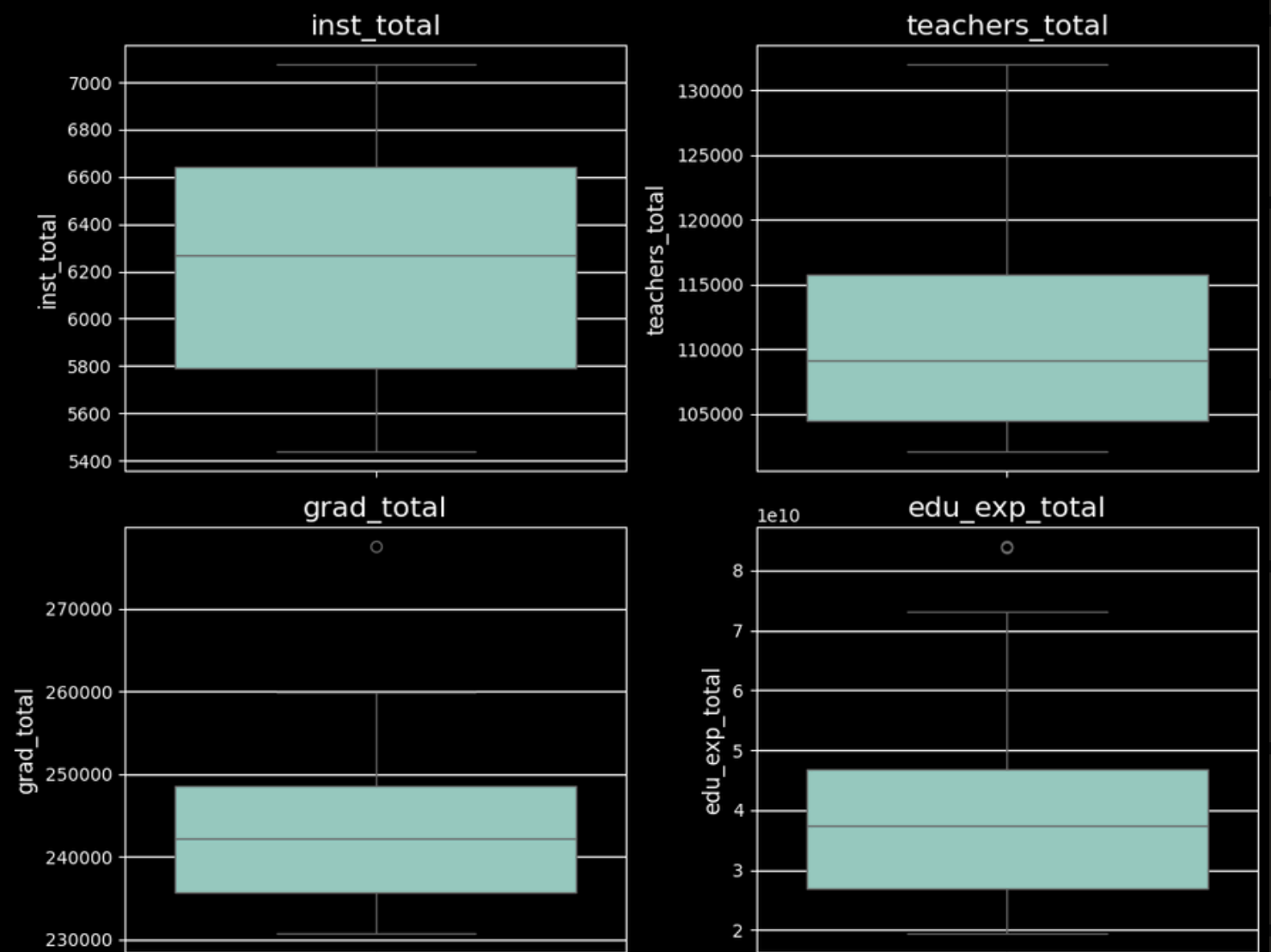


BOXPLOTS

Distribution of Economy Variables: Boxplots



Distribution of Education Variables: Boxplots



**Most of the distributions are
right-skewed.**

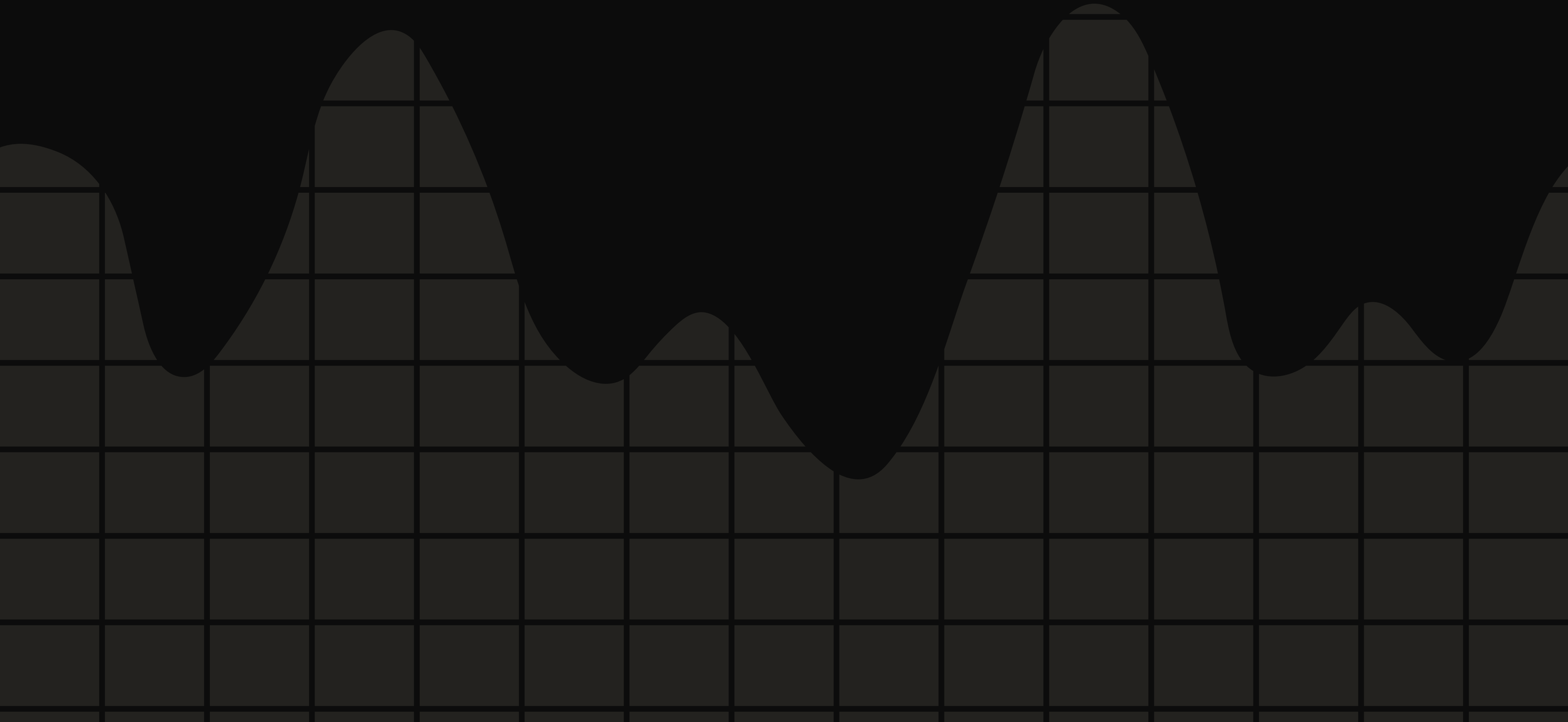
Exceptions:

- birth counts (normal — random influences),
- unemployment rate (left-skewed — ‘bad’ is in the other side)
- educational facilities (bimodal — state planning)

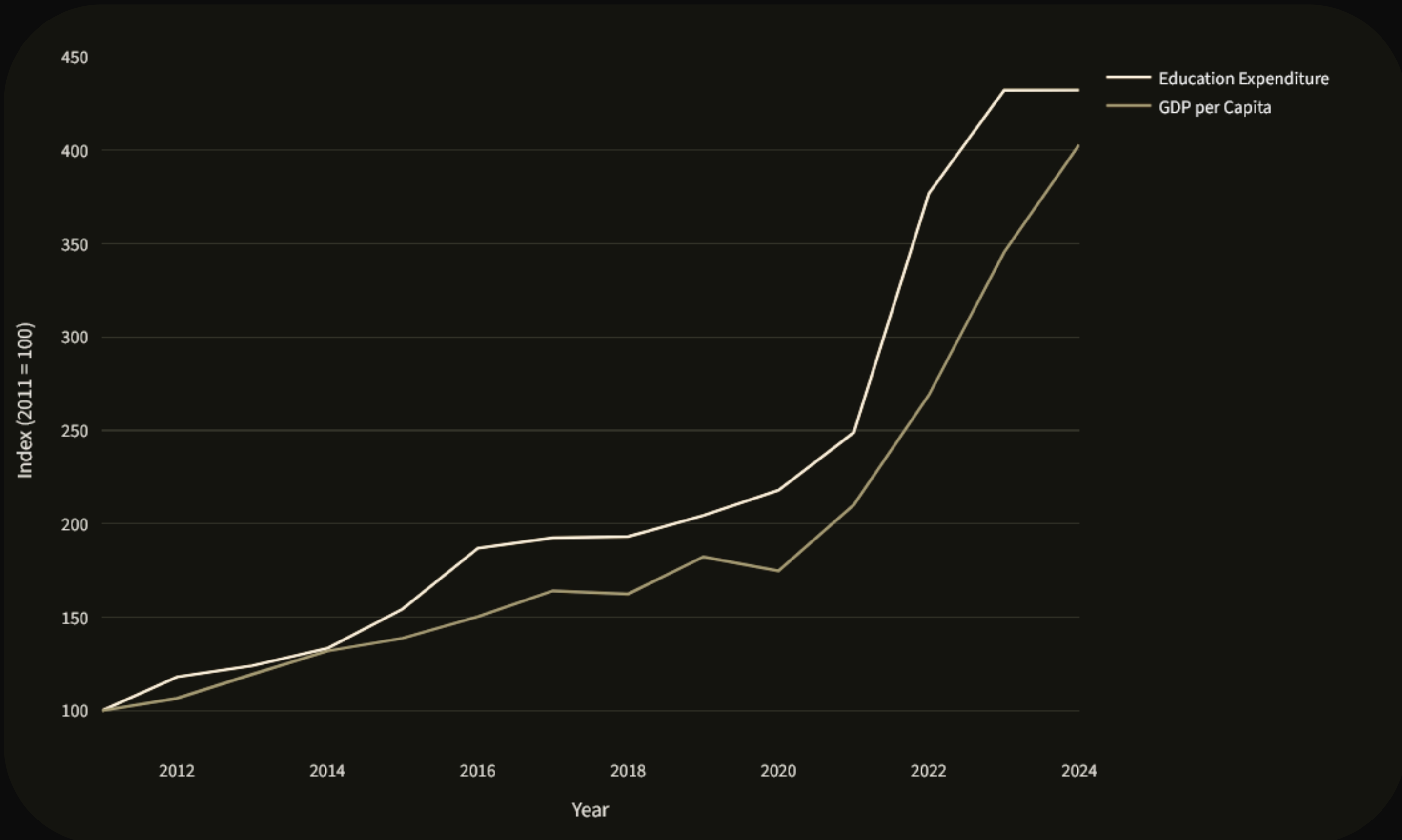
**GDP, salaries, education expenditure
and graduates got outliers**

- Recent spikes?
- Old records?

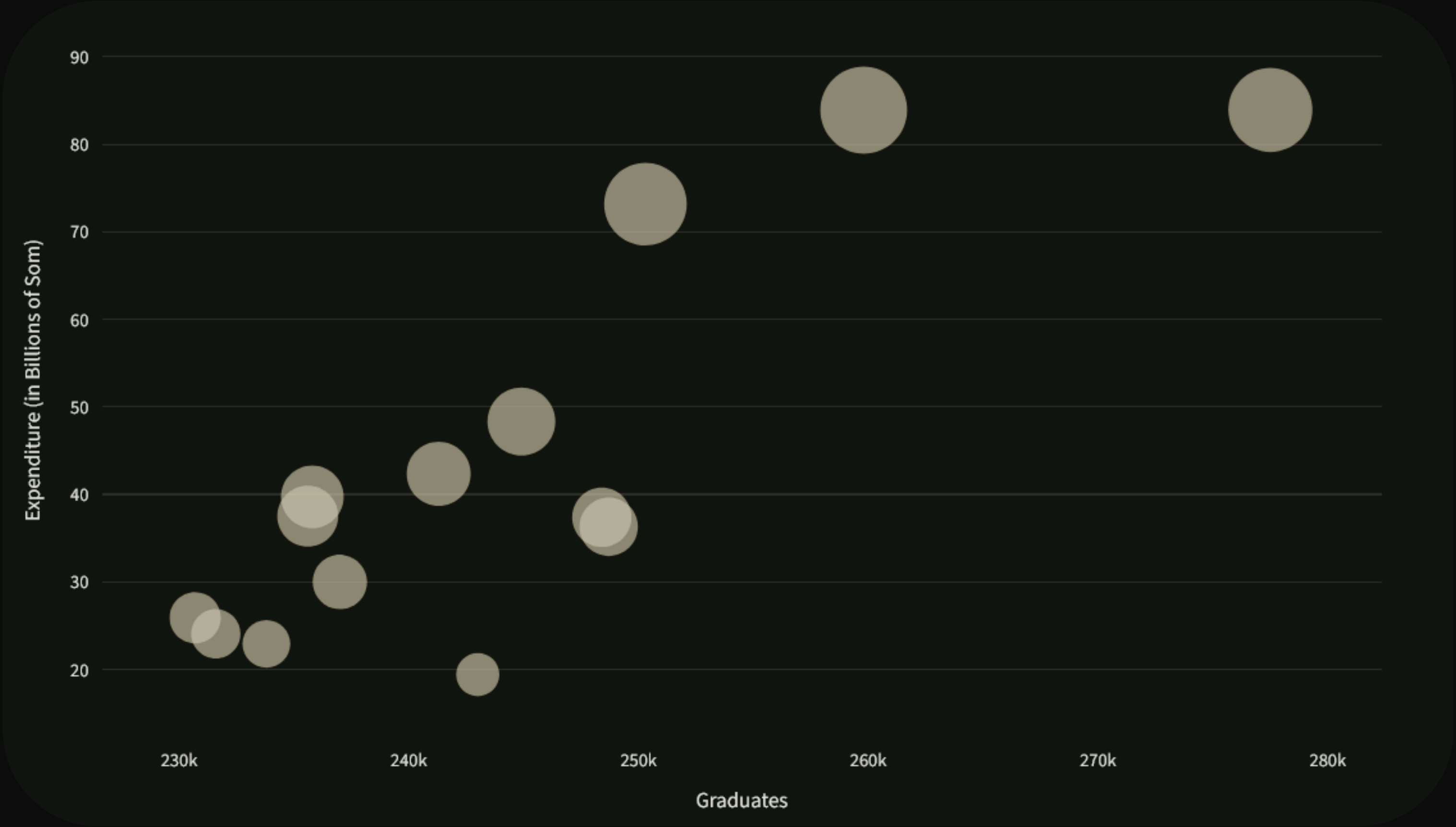
VISUALIZATIONS



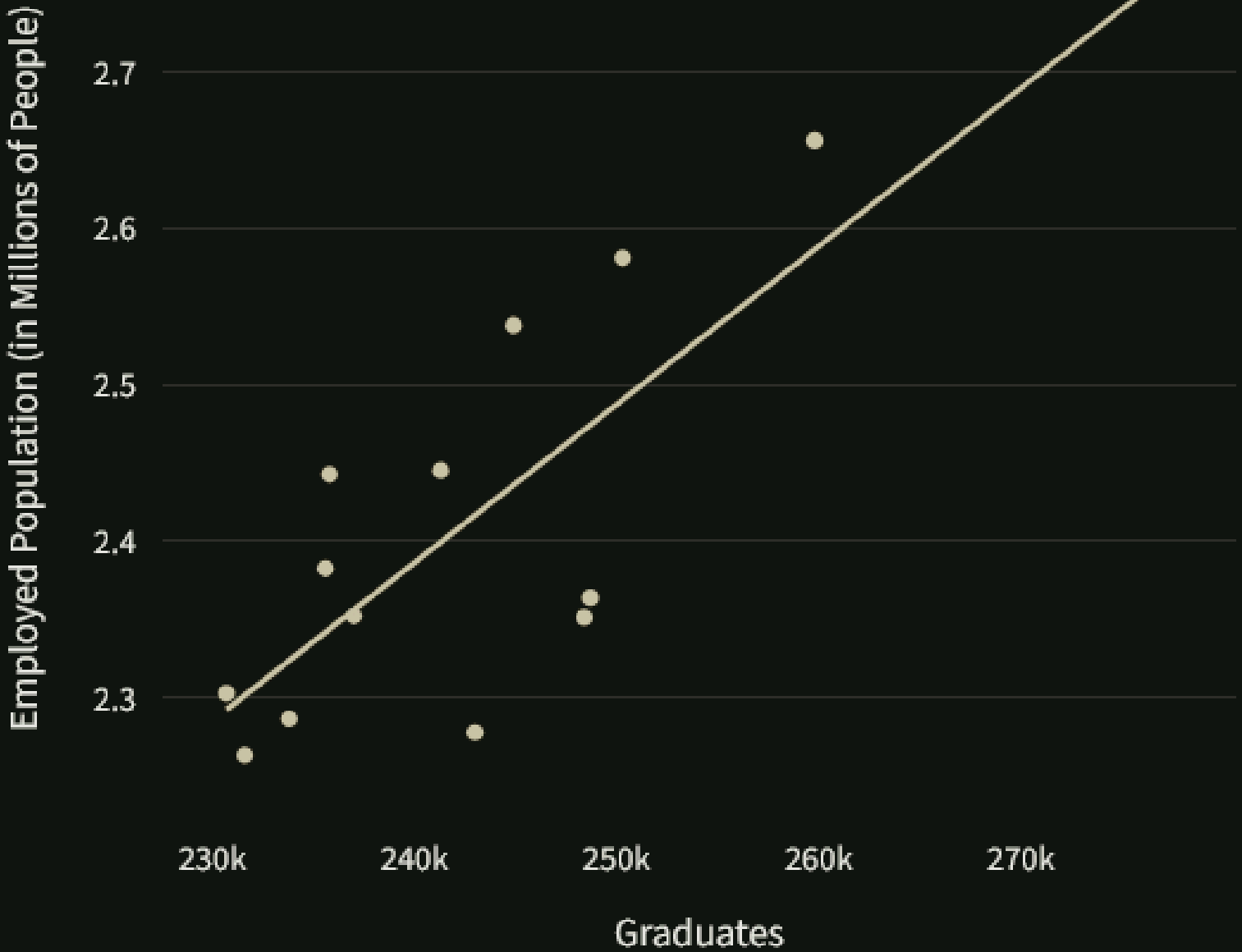
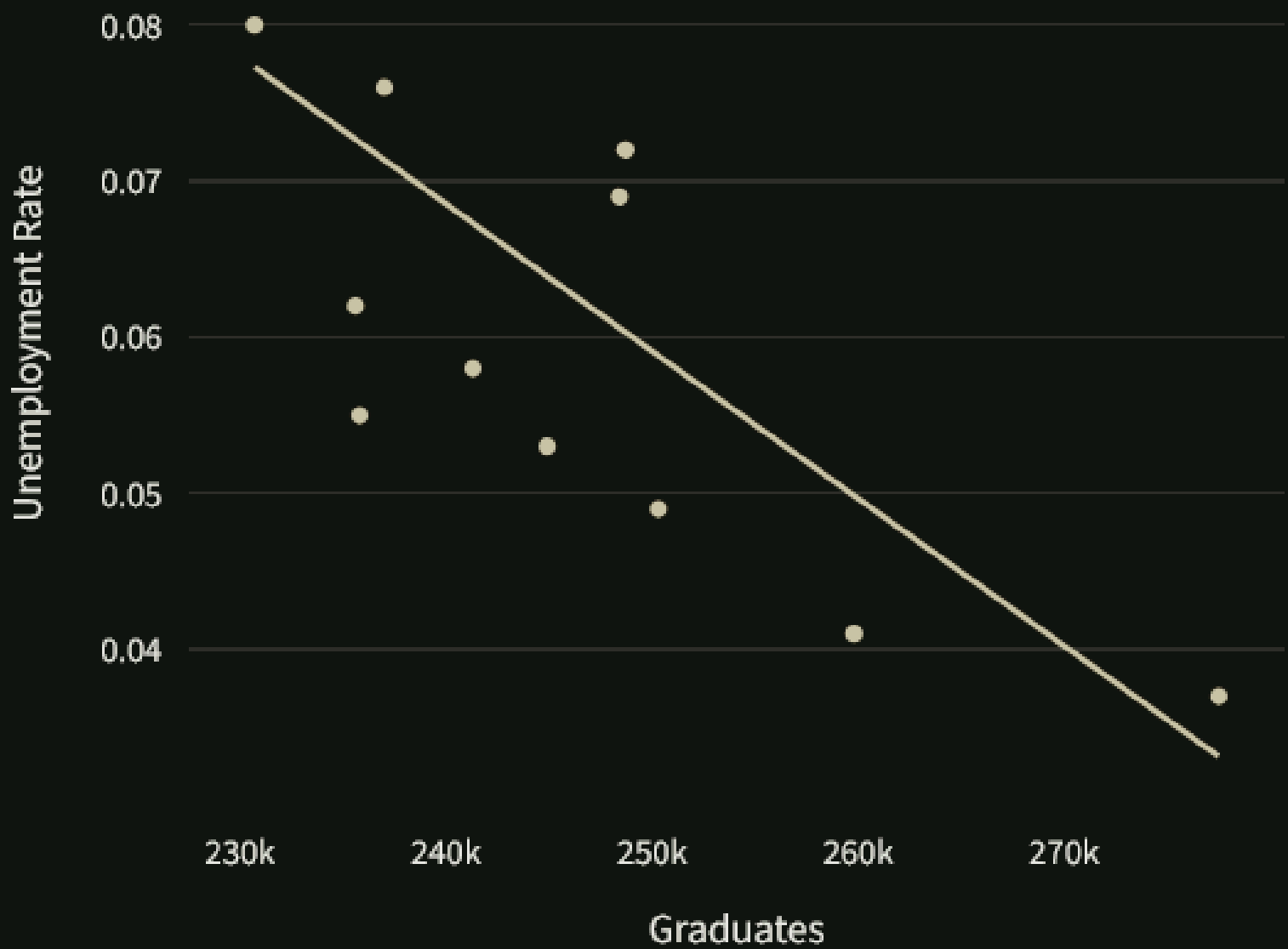
EDUCATION EXPENDITURE VS. GDP PER CAPITA (INDEXED)



EDUCATION EXPENDITURE VS. GRADUATES & COST PER GRADUATE



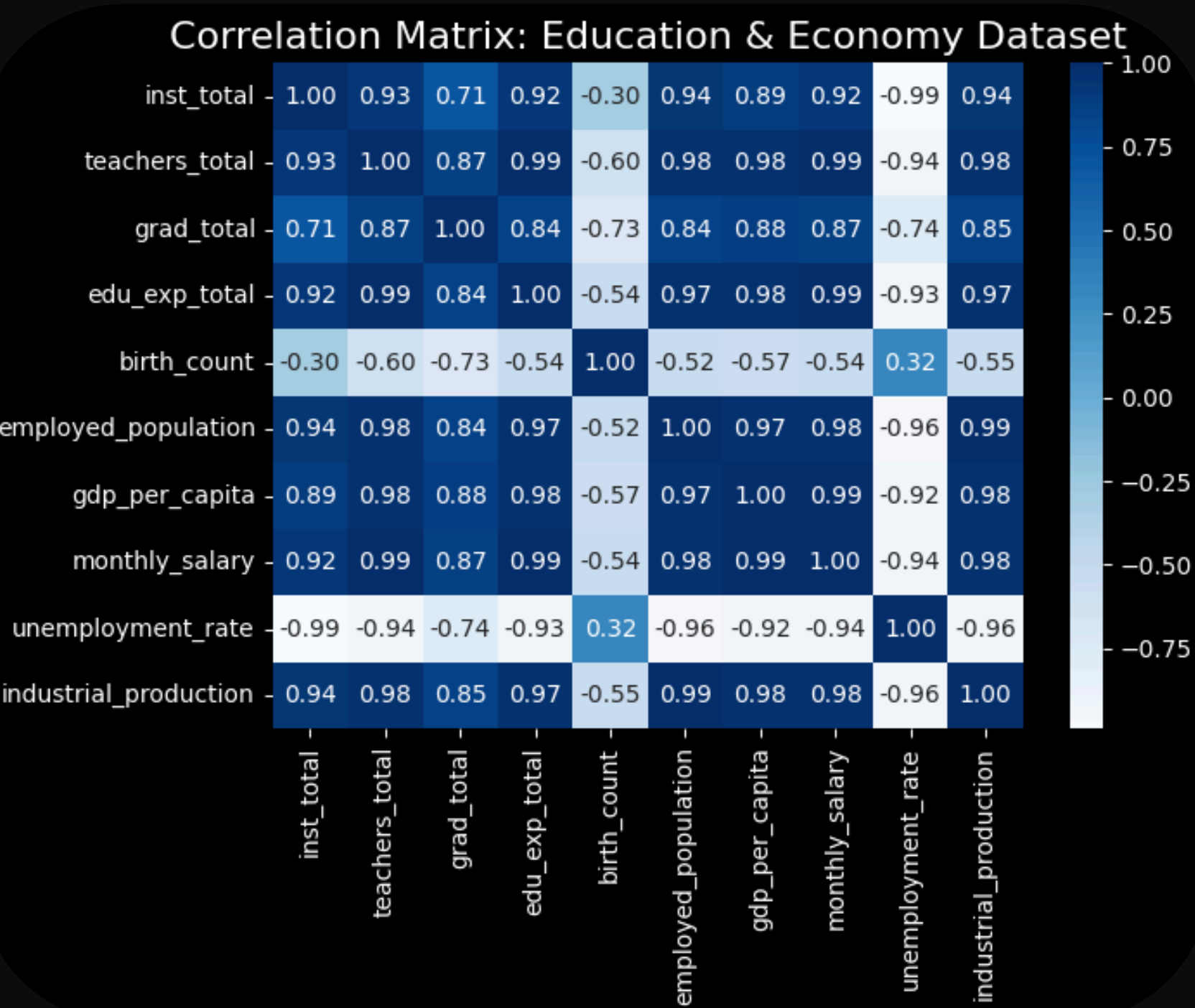
GRADUATES VS. UNEMPLOYMENT RATE & EMPLOYED POPULATION



CORRELATION ANALYSIS

Nearly all education and economy variables are strongly positively correlated with each other (0.84–0.99), suggesting they all grow together over time.

Birth counts and unemployment rate moved inversely to everything — marking strong negative correlation.



HYPOTHESIS TEST

Null Hypothesis: number of graduates and employed population are independent;

Hypothesis: number of graduates and employed population are associated.

METHOD: Pearson Correlation

<R: 0.8440,

P-VALUE: 0.0001>

IF P-VALUE < 0.05:

NULL-HYPOTHESIS **REJECTED**

INTERPRETATION

The growth of the economic and education indicators was simultaneous.

Moments when growth of one set of variables preceded the growth of the other were extremely rare.

Therefore, economy and education are *mutual accelerators*.



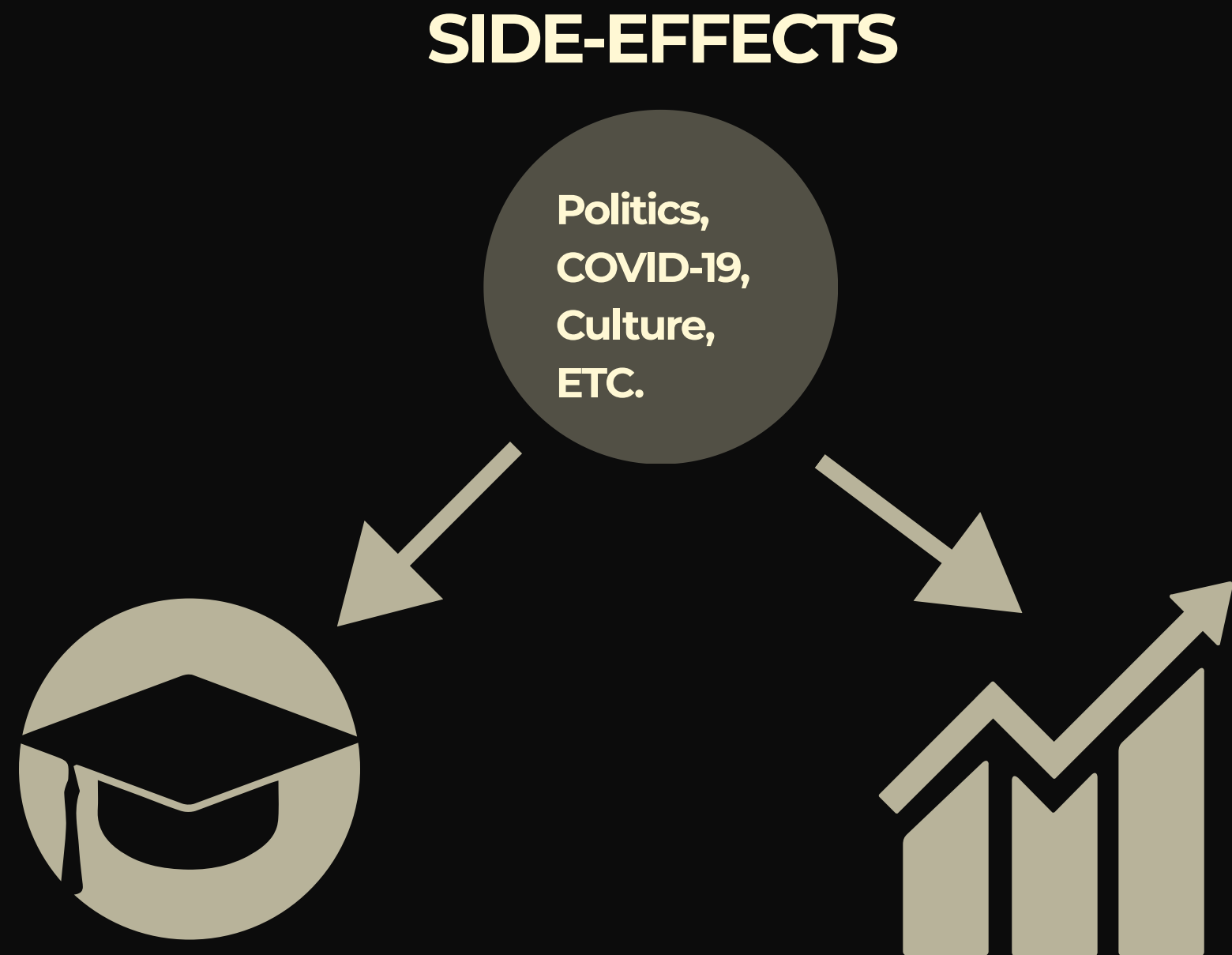
CONCLUSION & LIMITATIONS

Causation does *NOT* imply causality.

Are education and economics strongly correlated? *YES*.

Does growth in one of them causes growth in the other? *We do NOT know*.

Why? There are side-effects that are distorting the relationship, affecting both sides.



THANK YOU!

**A. TURDUEV,
R. MUKHAMMEDOV**